

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	04 July 2026
Team ID	
Project Name	Nutrition Assistant Using MERN stack
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a data analyst, I can collect the Agricultural Crop Production Dataset (1997–2021).	2	High	Entire Team
Sprint-1	Data Cleaning	USN-2	As a data analyst, I can clean missing, duplicate, and inconsistent records.	3	High	Entire Team
Sprint-1	Data Preparation	USN-3	As a data analyst, I can convert production units and validate data quality.	3	High	Entire Team
Sprint-2	Dashboard Design	USN-4	As a user, I can view KPI cards showing Total Production, Cultivated	3	High	Entire Team

			Area, and Yield.			
Sprint-2	Interactive Dashboard	USN-5	As a user, I can filter data by State, Crop, Year, and Season	5	High	Entire Team
Sprint-3	Visualizations	USN-6	As a user, I can analyze production trends using charts and maps	3	High	Entire Team
Sprint-3	Comparative Analysis	USN-7	As a researcher, I can compare crop production across states.	3	Medium	Entire Team
Sprint-3	Yield Analysis	USN-8	As a policymaker, I can identify high and low yield regions.	3	High	Entire Team
Sprint-4	Story Dashboard	USN-9	As a user, I can navigate through the Tableau Story for insights.	2	Medium	Entire Team
Sprint-4	Web Integration	USN-10	As a user, I can access the dashboard through a responsive website.	5	High	Entire Team
Sprint-4	Dashboard Publishing	USN-11	As a user, I can access the published Tableau Dashboard online.	2	High	Entire Team

Sprint-4	Documentation	USN-12	As a project reviewer, I can view the project documentation and GitHub repository.	2	Medium	Entire Team
----------	---------------	--------	--	---	--------	-------------

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	2 Days	30 June 2026	1 July 2026	20	2 July
Sprint-2	20	2 Days	02 July	3 July	20	4 July
Sprint-3	20	2 Days	5 July	6 July	20	6 July
Sprint-4	20	2 Days	6 July	7 July	20	7 July

Velocity:

Imagine we have a 7-day sprint duration, and the velocity of the team is 16 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \text{sprint duration} / \text{velocity} = 16 / 7 = 2.28$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

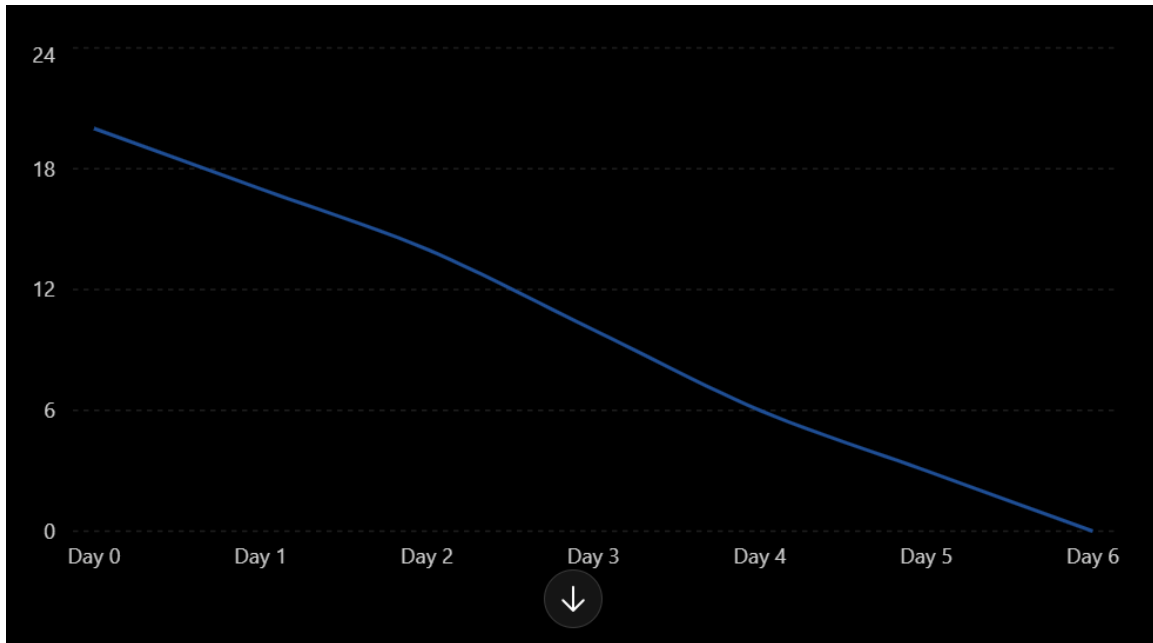


Figure: Sprint Burndown Chart for India's Agricultural Crop Production Analysis